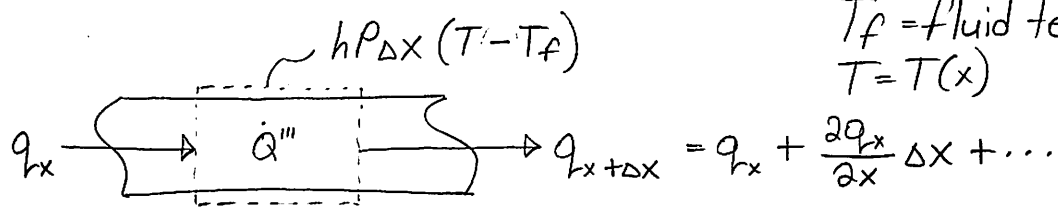


Fins with Heat Generation ($\dot{Q}'''$)



Writing out our energy balance

$$Aq_x - A\left(q_x + \frac{\partial q_x}{\partial x} \Delta x + \frac{1}{2} \frac{\partial^2 q_x}{\partial x^2} \Delta x^2 + \text{H.O.T.}\right) + \dot{Q}''' A \Delta x - hP \Delta x (T - T_f) = 0$$

We know from Fourier's Law: $q_x = -k \frac{\partial T}{\partial x}$

$$+kA \left(\frac{\partial^2 T}{\partial x^2} + \frac{1}{2} \frac{\partial^3 T}{\partial x^3} \Delta x + \text{H.O.T.} \right) + \dot{Q}''' A - hP (T - T_f) = 0$$

Letting $\Delta x \rightarrow 0$, and $\Theta = T - T_f$
 $d\Theta = dT$

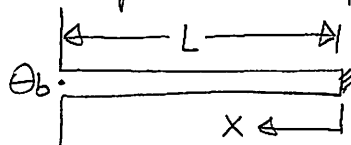
We obtain:

$$\frac{\partial^2 \Theta}{\partial x^2} + \frac{\dot{Q}'''}{k} - m^2 \Theta = 0, \quad m^2 = \frac{hP}{kA}$$

Assuming the insulated tip condition, our B.C.'s become

$$\left. \frac{\partial \Theta}{\partial x} \right|_{x=0} = 0$$

$$\Theta(x=L) = \Theta_b$$



To solve, we can do the particular and homogeneous solution:
Particular:

$$\frac{\partial^2 \Theta_p}{\partial x^2} + \frac{\dot{Q}'''}{k} - m^2 \Theta_p = 0$$

Set this to zero (quick solution for the particular)

$$\Theta_p = \frac{\dot{Q}'''}{km^2} = \frac{\dot{Q}''' kA}{k h P} = \frac{\dot{Q}''' A}{h P} \Rightarrow \boxed{\Theta_p = \frac{\dot{Q}''' A}{h P}}$$

Now we move on to the homogeneous part.

Homogeneous: $\frac{\partial^2 \Theta_h}{\partial x^2} - m^2 \Theta_h = 0$

We've already solved this problem before. Nothing but

$$\Theta_h = C_1 \cosh(mx) + C_2 \sinh(mx)$$

Now we combine $\Theta_p + \Theta_h = \Theta(x)$ and apply our B.C.'s

$$\Theta(x) = \Theta_p(x) + \Theta_h(x) = \frac{\dot{Q}''' A}{hP} + C_1 \cosh(mx) + C_2 \sinh(mx)$$

$$\left. \frac{\partial \Theta}{\partial x} \right|_{x=0} = 0 = -C_1 m \sinh(m(0)) + C_2 m \cosh(m(0)) \Rightarrow C_2 = 0$$

$$\Theta(x=L) = \Theta_b = \frac{\dot{Q}''' A}{hP} + C_1 \cosh(mL)$$

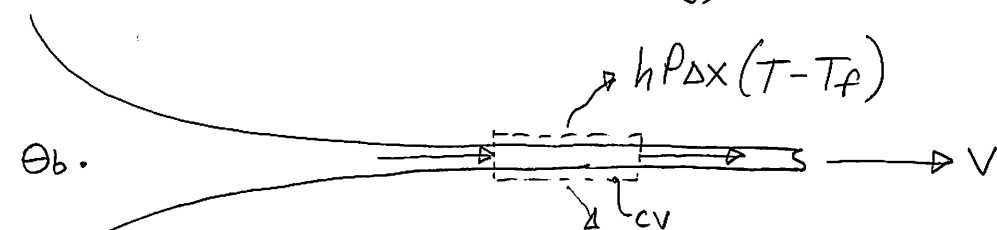
$$C_1 = \frac{\Theta_b - \frac{\dot{Q}''' A}{hP}}{\cosh(mL)}$$

$$\therefore \Theta(x) = \frac{\dot{Q}''' A}{hP} + \frac{\Theta_b - \frac{\dot{Q}''' A}{hP}}{\cosh(mL)} \cdot \cosh(mx)$$

$$\Theta(x) = \Theta_b \frac{\cosh(mx)}{\cosh(mL)} + \frac{\dot{Q}''' A}{hP} \left(1 - \frac{\cosh(mx)}{\cosh(mL)} \right)$$

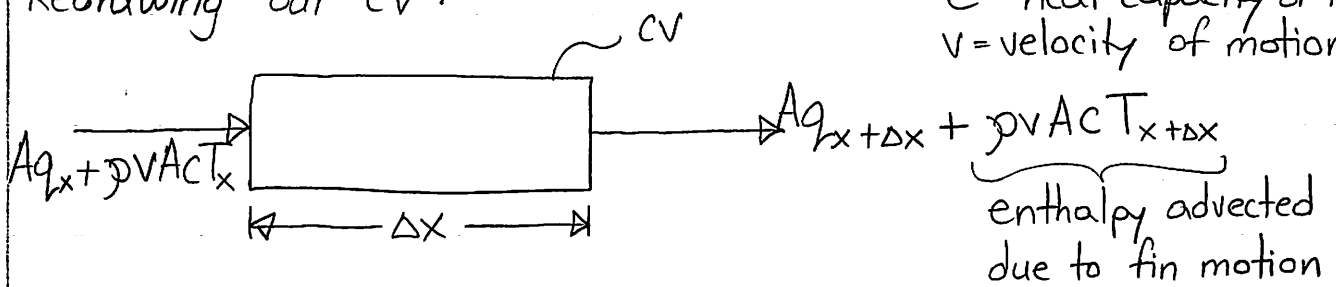
↳ Temperature profile in the fin with heat generat.

Moving Fins (For example, metal extrusion and solidification
⇒ continuous casting)



Our inlet and outlet terms become a bit more complicated.

Redrawing our CV:



Energy balance on our control volume:

$$Aq_x + \rho v A c T_x - (Aq_{x+\Delta x} + \rho v A c T_{x+\Delta x}) - h P \Delta x (T - T_f) = 0$$

Substituting in Fourier's Law and doing a Taylor series expans.

$$\begin{aligned}
 Aq_x + \rho v A c T_x - \left(Aq_x + A \frac{\partial q_x}{\partial x} \Delta x + A \frac{1}{2} \frac{\partial^2 q_x}{\partial x^2} \Delta x^2 + \text{H.O.T.} + \rho v A c T_x + \rho v A c \frac{\partial T}{\partial x} \Delta x + \rho v A c \frac{\partial^2 T}{\partial x^2} \frac{\Delta x^2}{2} + \text{H.O.T.} \right) - h P \Delta x (T - T_f) = 0 \\
 k A \frac{\partial^2 T}{\partial x^2} \Delta x + \frac{1}{2} k A \frac{\partial^3 T}{\partial x^3} \Delta x^2 + \text{H.O.T.} - \rho v A c \frac{\partial T}{\partial x} \Delta x - \rho v A c \frac{\partial^2 T}{\partial x^2} \frac{\Delta x^2}{2} + \text{H.O.T.} - h P \Delta x (T - T_f) = 0
 \end{aligned}$$

Let $\Delta x \rightarrow 0$, all H.O.T. terms drop out, we are left with:

$$k A \frac{\partial^2 T}{\partial x^2} - \rho v A c \frac{\partial T}{\partial x} - h P (T - T_f) = 0$$

$$\frac{\partial^2 T}{\partial x^2} - \frac{\rho v c}{k} \frac{\partial T}{\partial x} - \underbrace{\frac{h P}{k A}}_{m^2} (T - T_f) = 0$$

$$\text{Let } \Theta = T - T_f$$

$$\frac{d\Theta}{dx} = \frac{dT}{dx}$$

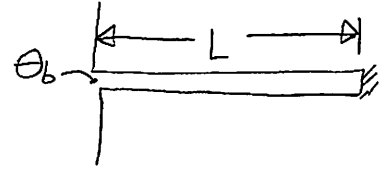
$$\frac{\partial^2 \Theta}{\partial x^2} - \frac{\rho v c}{k} \frac{\partial \Theta}{\partial x} - m^2 \Theta = 0$$

Our B.C.'s are (for infinite fin.) $\Rightarrow \begin{cases} (1) \Theta(x=0) = \Theta_b \\ (2) \Theta(x \rightarrow \infty) = 0 \end{cases}$

Note, if we non dimensionalize, it becomes more powerfull

$$\frac{\partial^2 \theta}{\partial x^2} - \frac{\rho v c}{k} \frac{\partial \theta}{\partial x} - m^2 \theta = 0$$

Let $\theta^* = \frac{\theta}{\theta_b}$, $x^* = \frac{x}{L}$



$$d\theta = \theta_b d\theta^*, \quad dx = L dx^* \Rightarrow \text{Back substituting}$$

$$\frac{\partial^2 (\theta_b \theta^*)}{L^2 \partial x^{*2}} - \frac{\rho v c}{k} \frac{\partial (\theta_b \theta^*)}{L \partial x^*} - m^2 (\theta_b \theta^*) = 0$$

$$\frac{1}{L^2} \frac{\partial^2 \theta^*}{\partial x^{*2}} - \frac{\rho v c}{k L} \frac{\partial \theta^*}{\partial x^*} - m^2 \theta^* = 0$$

Multiply through by L^2 on both sides

$$\frac{\partial^2 \theta^*}{\partial x^{*2}} - \underbrace{\frac{L \rho v c}{k}}_{\alpha} \frac{\partial \theta^*}{\partial x^*} - (mL)^2 \theta^* = 0$$

This is interesting: $\alpha = \frac{k}{\rho c}$

This becomes: $\frac{LV}{\alpha} \rightarrow \text{Thermal advection}$
 $\alpha \rightarrow \text{Thermal diffusion}$

Note, a side definition: Convection = motion of fluid in response to heat. (Diffusion + Advection)

Advection = motion of fluid, mass, momentum or heat due to the motion of the fluid itself.

Back to our problem:

$$Pe = \text{Peclet Number} = \frac{\text{advective transport rate}}{\text{diffusive transport rate}} = \frac{LV}{\alpha}$$

Lets first test the limits of our equation

$$\frac{LV}{\alpha} = \frac{\rho c L V}{k} = \frac{\rho c V}{\frac{k}{L}} \frac{\Delta T}{\Delta T} = \frac{\rho c V \Delta T}{\frac{k \Delta T}{L}} \rightarrow \frac{q''_{adv}}{q''_{cond}} \quad (57)$$

If advection dominates, $\frac{LV}{\alpha} \gg 1$

Our diffusion term drops out.

$$\frac{\partial^2 \theta}{\partial x^2} - \frac{\rho V C}{k} \frac{\partial \theta}{\partial x} - m^2 \theta = 0$$

0 (negligible)

$$\frac{V}{\alpha} \frac{\partial \theta}{\partial x} + m^2 \theta = 0 \quad (\text{Rearranging})$$

$$\frac{1}{\theta} \frac{\partial \theta}{\partial x} = - \frac{m^2 \alpha}{V} = - \frac{h P K}{K A V \rho C} = - \frac{h P}{\rho C V A}$$

$$\int \frac{d\theta}{\theta} = - \frac{h P}{\rho C V A} \int dx$$

$$\ln(\theta) = - \frac{h P}{\rho C V A} x + C_1$$

Constant of integration, not heat capacity C.

$$e^{\ln(\theta)} = e^{-\frac{h P}{\rho C V A} x + C_1} = C_1' e^{-\frac{h P}{\rho C V A} x}$$

$$\theta(x) = C_1' e^{-\frac{h P x}{\rho C V A}}$$

We know that $\theta(x=0) = \theta_b = C_1' e^0 \Rightarrow C_1' = \theta_b$

Let's check our second b.c.

$$\theta(x \rightarrow \infty) = 0 \Rightarrow \theta_b e^{-\infty} \rightarrow 0 \quad \text{OK!}$$

$$\therefore \boxed{\theta(x) = \theta_b e^{-\frac{h P x}{\rho C V A}}} \Rightarrow \text{When advection dominates diffusion.}$$

But what if conduction is non-negligible? (i.e. $Pe \sim 1$)

Now we have to solve the full differential equation

Start with the characteristic equation:

$$\text{Let } \theta = e^{\lambda x}$$

$$\frac{\partial \theta}{\partial x} = \lambda e^{\lambda x}$$

$$\frac{\partial^2 \theta}{\partial x^2} = \lambda^2 e^{\lambda x}$$

} Substitute into our ODE

$$\lambda^2 e^{\lambda x} - \frac{V}{\alpha} \lambda e^{\lambda x} - m^2 e^{\lambda x} = 0$$

$$\lambda^2 - \frac{V}{\alpha} \lambda - m^2 = 0 \quad (\text{Quadratic equation})$$

$$\lambda = \frac{V}{2\alpha} - \sqrt{\frac{V^2}{4\alpha^2} + m^2} \Rightarrow \text{Only use negative root since we need } \Theta(x \rightarrow \infty) = 0$$

$$\lambda = -\frac{V}{2\alpha} \left[\sqrt{1 + \frac{4m^2\alpha^2}{V^2}} - 1 \right] \Rightarrow \Theta = \Theta_b e^{\lambda x}$$

To check our solution, if $\left(\frac{\alpha}{V}\right)^2 \ll 1$ (Advection Dominates)

Then $\frac{4m^2\alpha^2}{V^2} \ll 1$, we can do a Taylor series expansion

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{4} + \frac{3x^3}{8} - \dots$$

$$\begin{aligned} f(x) &= (1+x)^{1/2} \\ \frac{\partial f}{\partial x} &= \frac{1}{2}(1+x)^{-1/2} \\ \frac{\partial^2 f}{\partial x^2} &= -\frac{1}{4}(1+x)^{-3/2} \\ &\vdots \end{aligned}$$

So our solution becomes: if $x \ll 1$. H.O.T. can be neglected

$$\lambda = -\frac{V}{2\alpha} \left[\sqrt{1 + \frac{4m^2\alpha^2}{V^2}} - 1 \right]$$

$$= -\frac{V}{2\alpha} \left(1 + \frac{2m^2\alpha^2}{V^2} - 1 \right) = -\frac{m^2\alpha}{V}$$

$$\Theta = \Theta_b e^{\left(-\frac{hPx}{kAV}\right)x}$$

$$\Theta = \Theta_b e^{-\frac{hPx}{P_{CAV}}} \Rightarrow \text{Matches our previous solution}$$

So what if we move with our element and map out our temperature with respect to time.



$$\theta = \theta_b e^{-\frac{h\rho x}{\rho c A V}} \Rightarrow \text{Note, } t = \frac{x}{V}$$

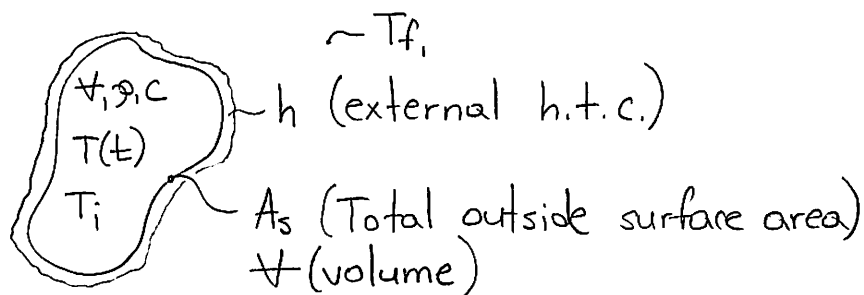
$$\theta = \theta_b e^{-\frac{h\rho}{\rho c A} t} \Rightarrow \text{Multiply by } \frac{\Delta x}{\Delta x} \text{ in the exponent}$$

$$\theta = \theta_b e^{-\frac{h(\rho \Delta x)}{\rho c (A \Delta x)} t} \Rightarrow \rho \Delta x = A_s \quad (\text{Total outside surface area})$$

$$A \Delta x = V \quad (\text{Total fin volume in element})$$

$$\boxed{\theta = \theta_b e^{-\frac{h A_s}{\rho c V} t}} \Rightarrow \text{Looks familiar (Lumped capacitance model)}$$

Lumped Capacitance Why do we care? Forensic analysis of dead bodies.



Energy ballance: $\dot{E}_{\text{stored}} + \dot{E}_{\text{in}} - \dot{E}_{\text{out}} + \dot{E}_{\text{gen}} = 0$

$$\frac{\partial}{\partial t} (V \rho c T) = h A_s (T - T_f)$$

$$V \rho c \frac{\partial T}{\partial t} - h A_s (T - T_f) = 0$$

$$\frac{\partial T}{\partial t} - \frac{h A_s}{V \rho c} (T - T_f) = 0 \Rightarrow \text{Let } \theta = T - T_f$$

$$\frac{\partial \theta}{\partial t} - \frac{h A_s}{V \rho c} \theta = 0 \Rightarrow \text{Let } \lambda = \frac{h A_s}{V \rho c}$$

$$\frac{\partial \theta}{\partial t} - \lambda \theta = 0 \Rightarrow \int \frac{\partial \theta}{\theta} = \int \lambda dt$$

$$\ln \theta + \ln C_1 = \lambda t + C_2 \Rightarrow \text{Combine constants}$$

$$\ln \theta = \lambda t + C_3$$

$$e^{\ln \theta} = e^{\lambda t + C_3} \Rightarrow \theta = C_4 e^{\lambda t}$$

Our B.C. is $\Theta(t=0) = \Theta_i = T_i - T_f$

$$\Theta(t=0) = C_4 e^{-\lambda \cdot 0} = \Theta_i \Rightarrow C_4 = \Theta_i$$

$$\therefore \boxed{\Theta = \Theta_i e^{-\lambda t} = \Theta_i e^{-\frac{h A_s t}{\rho c}}} \Rightarrow \text{Lumped capacitance model.}$$

Note, only valid if the whole body is isothermal; $\frac{h t}{k} < 0.1$

Non-constant Fluid Temperature ($T(t)$ = body temperature)
Our previous equation would become: ($T_f(t)$ = fluid temp.)

$$\left[\frac{\partial T}{\partial t} + \lambda T = \lambda T_f(t) \right] \cdot e^{\lambda t} \quad (\text{Multiply by } e^{\lambda t}, \text{ integrating factor method})$$

$$\underbrace{e^{\lambda t} \frac{\partial T}{\partial t} + \lambda e^{\lambda t} T}_{\frac{\partial}{\partial t} (e^{\lambda t} T)} = \lambda e^{\lambda t} T_f(t)$$

$$\frac{\partial}{\partial t} (e^{\lambda t} T) = \lambda e^{\lambda t} T_f(t) \quad (\text{Integrate both sides})$$

$$\int_0^t \frac{\partial}{\partial t} (e^{\lambda t} T) dt = \lambda \int_0^t e^{\lambda t} T_f(t) dt$$

$$\boxed{e^{\lambda t} T \Big|_0^t = \lambda \int_0^t e^{\lambda t} T_f(t) dt} \Rightarrow \text{General solution, } \lambda = \frac{h A_s}{\rho c}$$

Example | Let $T_f = \beta \cdot t$ where $\beta = \text{constant}$

I.C. $\Rightarrow t=0, T_f = T = 0$

$$e^{\lambda t} T \Big|_0^t = \lambda \int_0^t e^{\lambda t} (\beta t) dt$$

$$e^{\lambda t} T(t) - e^{\lambda \cdot 0} T(0) = e^{\lambda t} T(t) = \lambda \beta \int_0^t t e^{\lambda t} dt$$

So how do we solve this?

Let $s = \lambda t$, $ds = \lambda dt$

$$e^{\lambda t} T(t) = \cancel{\lambda} B \int_0^{\frac{s}{\lambda}} \frac{s}{\cancel{\lambda}} e^s \frac{ds}{\lambda}$$

$$= \frac{B}{\lambda} \int_0^{\frac{s}{\lambda}} s e^s ds = \frac{B}{\lambda} (s e^s - e^s) \Big|_0^{\frac{s}{\lambda}}$$

$$e^{\lambda t} T(t) = \frac{B}{\lambda} e^{\lambda t} (\lambda t - 1) \Big|_0^t$$

$$e^{\lambda t} T(t) = \frac{B}{\lambda} [e^{\lambda t} (\lambda t - 1) - e^0 (0 - 1)]$$

$$= \frac{B}{\lambda} [e^{\lambda t} (\lambda t - 1) + 1]$$

Dividing through by $e^{\lambda t}$

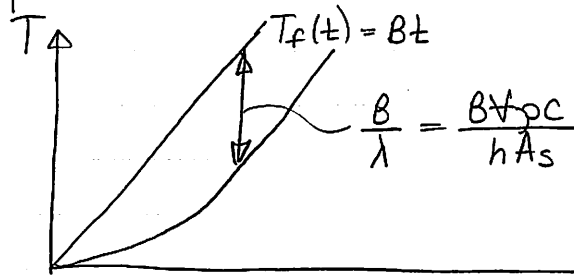
$$T(t) = \frac{B}{\lambda} [(\lambda t - 1) + e^{-\lambda t}]$$

$$= \underbrace{Bt}_{T_f(t)} - \frac{B}{\lambda} (1 - e^{-\lambda t})$$

$$T(t) = T_f(t) - \frac{B}{\lambda} (1 - e^{-\lambda t}) ; \quad \lambda = \frac{h A_s}{V \rho C}$$

Lag term. This is why its difficult to measure temperature changes in a fluid since measurement lags if B is high.

If we plot our result:



For a thermocouple:

$$\frac{B}{\lambda} = \frac{B V \rho C}{h A_s} = \frac{B R^2 \rho C}{h R^2}$$

$$= \frac{B R^2 \rho C}{h} \Rightarrow R \sim 0.001$$

$$h \sim 2500$$

$$\rho \sim 1000$$

$$C \sim 150$$

$$B \sim 2 \quad (62)$$

$$\frac{B}{\lambda} = \frac{2}{25} ^\circ\text{C}$$

↳ For Water w/ TC.

Example | Time of death.

Person is found dead at 5 PM in a room with $T_f = 20^\circ\text{C}$. The body temperature is $T = 25^\circ\text{C}$ when found.

$$h = 8 \text{ W/m}^2 \cdot \text{K}$$

Geometry: $D \approx 30 \text{ cm}$, $L = 1.7 \text{ m}$

Estimate the time of death.

A healthy persons temperature is $T_i = 37^\circ\text{C}$

Assume radiation is negligible: Check:

$$\begin{aligned} h_{\text{rad}} &= \epsilon \sigma (T_s^2 + T_\infty^2)(T_s + T_\infty) \\ &= (1.0)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)((37+273)^2 + (20+273)^2)(37+273 + 20+273) \\ h_{\text{rad}} &= 6.26 \text{ W/m}^2 \cdot \text{K} \end{aligned}$$

Add another $\bar{h}_{\text{rad}} \approx 5 \text{ W/m}^2 \cdot \text{K}$ to our convection h.t.c.

$$h_{\text{TOT}} = h_c + h_{\text{rad}} \approx 13 \text{ W/m}^2 \cdot \text{K}$$

Now lets assume a lumped capacitance model.

$$L_c = \frac{V}{A_s} = \frac{\pi r_o^2 L}{2\pi r_o L + 2\pi r_o^2} = 0.069 \text{ m}$$

$$\text{Check the Biot number: } Bi_{Lc} = \frac{h L_c}{k}$$

Assume $k = k_{\text{water}} = 0.6 \text{ W/m} \cdot \text{K}$ (since body is made up of 72% water by mass)

$$Bi_{Lc} \approx \frac{(13 \text{ W/m}^2 \cdot \text{K})(0.069 \text{ m})}{0.6 \text{ W/m} \cdot \text{K}} = 1.5 > 0.1$$

So lumped capacitance is not applicable but we use it to get a rough estimate:

$$\lambda = \frac{h A_s}{\rho C V} = \frac{h}{\rho C L_c} = 4.53 \times 10^{-5} \text{ s}^{-1} \Rightarrow \frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-\lambda t} \Rightarrow t = 7.5 \text{ hr}$$